Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; gemma-4-26b-a4b

Abstract

This report evaluates `google/gemma-4-26b-a4b` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: OpenRouter leads Cache hit rate in all routing modes; OpenRouter leads Observed cost in all routing modes; OpenRouter leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes.

Overall, Infron does not show a clear cross-mode strength in this run, while OpenRouter's cross-mode strengths are throughput, Streaming TTFT, E2E latency, observed cost, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform-level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput	Observed cost	E2E latency	Streaming TTFT	Cache hit rate
OpenRouter wins 4/4 Max advantage 101.85%, higher is better	OpenRouter wins 4/4 Max advantage 99.04%, lower is better	OpenRouter wins 4/4 Max advantage 101.87%, lower is better	OpenRouter wins 4/4 Max advantage 103.64%, lower is better	OpenRouter wins 4/4 Max advantage 50.70%, higher is better

Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 15.57%	OpenRouter advantage 31.29%	OpenRouter advantage 15.58%	OpenRouter advantage 11.60%	OpenRouter advantage 0.00%
Price First price	OpenRouter advantage 28.16%	OpenRouter advantage 36.73%	OpenRouter advantage 28.16%	OpenRouter advantage 23.13%	OpenRouter advantage 50.70%
Latency First latency	OpenRouter advantage 76.57%	OpenRouter advantage 99.04%	OpenRouter advantage 76.64%	OpenRouter advantage 78.36%	OpenRouter advantage 68.15%
TTFT First ttft	OpenRouter advantage 101.85%	OpenRouter advantage 24.59%	OpenRouter advantage 101.87%	OpenRouter advantage 103.64%	OpenRouter advantage 4.11%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second `usage.prompt_tokens` deltas are limited to 50 tokens within each `sort/group/round` pair.	Methods and data quality
Cache reuse	OpenRouter leads Cache hit rate in all routing modes	Overall metrics and mechanism section
Observed cost	OpenRouter leads Observed cost in all routing modes	Overall metrics and provider drill-down
Performance	OpenRouter leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)
Latency First	Minimize full-response waiting time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)
TTFT First	Minimize streaming first-token time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference-platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first-token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token-level cache hit rate for repeated stable prefixes.	Second-request cache-read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response-returned `usage.prompt_tokens` as the input-token control while allowing small cross-platform accounting variance up to 50 tokens.
- Extends prompt-caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

2. Experimental Design, Dataset, and Controls

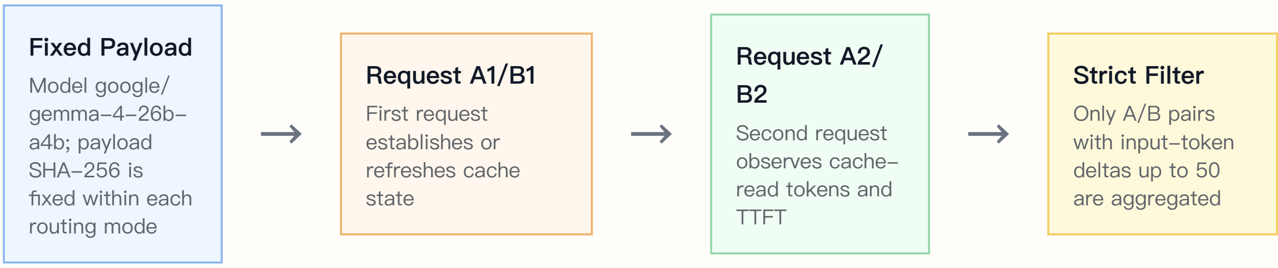
2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables

Figure 1: Experimental Design and Strict A/B Pairing Filter



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with `cache_read_tokens > 0`	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	`google/gemma-4-26b-a4b`
Provider model IDs	infron: `google/gemma-4-26b-a4b`; openrouter: `google/gemma-4-26b-a4b-it` (provider alias mapping recorded)
Platforms	Infron and OpenRouter
API protocol	`/v1/chat/completions`
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	`short` ≈1500, `medium` ≈8000, `long` ≈32000
Excluded records	108

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	186	7013376	0.00%	$0.49143800	1.14 tok/s	3518.51 ms	3198.14 ms
Throughput First	**OpenRouter**	186	7013004	**99.36%**	**$0.37430218**	**1.31 tok/s**	**3044.12 ms**	**2865.78 ms**
Price First	Infron	165	5938290	1.81%	$0.42155500	0.93 tok/s	4283.03 ms	3875.66 ms
Price First	**OpenRouter**	165	5937978	**93.45%**	**$0.30830935**	**1.20 tok/s**	**3342.00 ms**	**3147.68 ms**
Latency First	Infron	195	7187631	51.00%	$0.91559200	0.73 tok/s	5452.12 ms	5187.90 ms
Latency First	**OpenRouter**	195	7184700	**85.75%**	**$0.46001202**	**1.30 tok/s**	**3086.54 ms**	**2908.68 ms**
TTFT First	Infron	200	7407473	81.56%	$0.62097300	0.70 tok/s	5707.61 ms	5390.37 ms
TTFT First	**OpenRouter**	200	7404114	**84.91%**	**$0.49840228**	**1.42 tok/s**	**2827.30 ms**	**2647.02 ms**

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	**2537.70 ms**	10988.81 ms	17290.74 ms	**2252.29 ms**	10797.79 ms	16680.14 ms
Throughput First	**OpenRouter**	2571.20 ms	**5819.18 ms**	**13868.07 ms**	2417.63 ms	**5716.57 ms**	**13614.74 ms**
Price First	Infron	3409.92 ms	9601.64 ms	21455.50 ms	2910.42 ms	9373.30 ms	20757.24 ms
Price First	**OpenRouter**	**2546.55 ms**	**8403.99 ms**	**16002.68 ms**	**2401.96 ms**	**8337.77 ms**	**15772.69 ms**
Latency First	Infron	4471.93 ms	10126.98 ms	17210.69 ms	4254.40 ms	9630.82 ms	17007.03 ms
Latency First	**OpenRouter**	**2482.57 ms**	**5164.90 ms**	**8160.89 ms**	**2221.57 ms**	**4925.94 ms**	**8140.31 ms**
TTFT First	Infron	4817.81 ms	11790.30 ms	18591.58 ms	4424.24 ms	11326.14 ms	18067.71 ms
TTFT First	**OpenRouter**	**2327.37 ms**	**5387.84 ms**	**11818.53 ms**	**2151.59 ms**	**5242.32 ms**	**11016.61 ms**

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	**−948.78 ms**	−1810.24 ms to −125.80 ms	0.0242	186	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	**−664.72 ms**	−1522.67 ms to 122.59 ms	0.1115	186	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	**−0.0611 tok/s**	−0.1694 tok/s to 0.0458 tok/s	0.2652	186	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	**$−0.00062976**	$−0.00074307 to $−0.00051465	<0.001	186	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	**−98.13 pp**	−99.24 pp to −96.51 pp	<0.001	186	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	**−1882.06 ms**	−2865.51 ms to −901.88 ms	<0.001	165	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	**−1455.95 ms**	−2437.73 ms to −481.50 ms	0.0027	165	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	**−0.3721 tok/s**	−0.4702 tok/s to −0.2626 tok/s	<0.001	165	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	**$−0.00068634**	$−0.00081721 to $−0.00056262	<0.001	165	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	**−89.57 pp**	−93.78 pp to −84.76 pp	<0.001	165	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	**−4731.16 ms**	−6219.92 ms to −3184.34 ms	<0.001	195	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	**−4558.45 ms**	−6074.58 ms to −2988.64 ms	<0.001	195	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p–value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	**–0.7792 tok/s**	–0.8619 tok/s to –0.6932 tok/s	<0.001	195	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	**$–0.00233631**	$–0.00272666 to $–0.00195797	<0.001	195	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	**–30.61 pp**	–38.88 pp to –21.65 pp	<0.001	195	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	**–5760.62 ms**	–6875.31 ms to –4760.89 ms	<0.001	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	**–5486.69 ms**	–6570.88 ms to –4505.71 ms	<0.001	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	**–0.8170 tok/s**	–0.9051 tok/s to –0.7328 tok/s	<0.001	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	**$–0.00061285**	$–0.00088470 to $–0.00033943	<0.001	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	**–6.23 pp**	–13.66 pp to 1.21 pp	0.1312	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	0	0.0000	0	0.00 ms	3198.14 ms	3518.51 ms
Throughput First	**OpenRouter**	0	0.0000	0	0.00 ms	**2865.78 ms**	**3044.12 ms**
Price First	Infron	0	0.0000	0	0.00 ms	3875.66 ms	4283.03 ms
Price First	**OpenRouter**	0	0.0000	0	0.00 ms	**3147.68 ms**	**3342.00 ms**
Latency First	Infron	0	0.0000	0	0.00 ms	5187.90 ms	5452.12 ms
Latency First	**OpenRouter**	0	0.0000	0	0.00 ms	**2908.68 ms**	**3086.54 ms**
TTFT First	Infron	0	0.0000	0	0.00 ms	5390.37 ms	5707.61 ms
TTFT First	**OpenRouter**	0	0.0000	0	0.00 ms	**2647.02 ms**	**2827.30 ms**

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions`; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
`chat_completions`	`/v1/chat/completions`	Infron	800	1600	**98.56%**	99.94%	99.94%	99.94%	**100.00%**
`chat_completions`	`/v1/chat/completions`	OpenRouter	800	1600	97.75%	**100.00%**	**100.00%**	**100.00%**	**100.00%**

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

Figure 3-E2: Normalized capability contour

Five metrics are normalized to 0-100, where higher is better.

Figure 3-E3: Cost-cache efficiency plane

X-axis is token cache hit rate; Y-axis is observed total cost.

Figure 3-E1: Routing-mode core metric comparison

Infron and OpenRouter are shown side by side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache-domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E3: Latency First core metric comparison

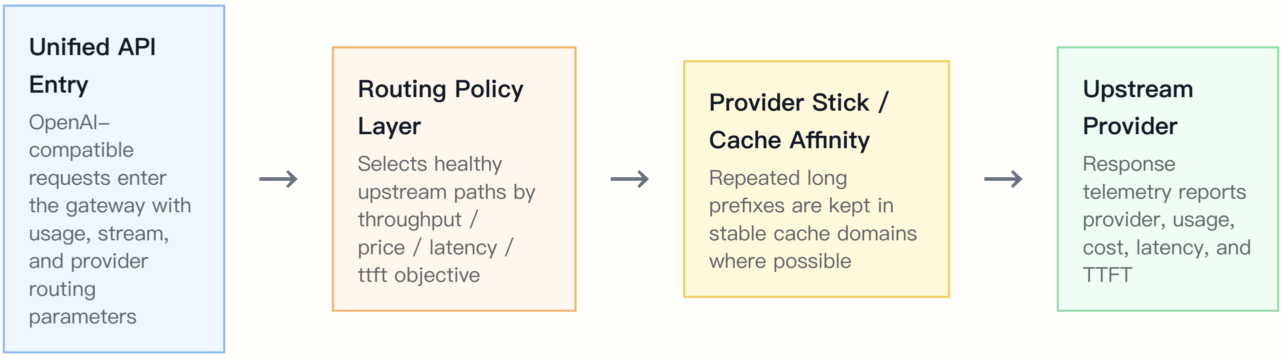
Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism

Figure 12: Infron Multi-Provider Routing and Cache Control Plane



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	372	372	`deepinfra` 372

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	372	372	`Wafer` 369, `Google` 3
Price First	Infron	330	330	`deepinfra` 324, `parasail` 6
Price First	OpenRouter	330	330	`Wafer` 312, `DeepInfra` 7, `SiliconFlow` 7, `DekaLLM` 4
Latency First	Infron	390	390	`novita` 326, `nextbit` 64
Latency First	OpenRouter	390	390	`Wafer` 331, `Google` 59
TTFT First	Infron	400	400	`nextbit` 267, `parasail` 78, `deepinfra` 53, `novita` 2
TTFT First	OpenRouter	400	400	`Wafer` 327, `Google` 73

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/ second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	`deepinfra`	372	100.00%	186/186	186	3198.14 ms	3518.51 ms	7013376	1488
Throughput First	OpenRouter	`Wafer`	369	99.19%	184/185	186	2869.08 ms	3047.20 ms	6980589	1476
Throughput First	OpenRouter	`Google`	3	0.81%	2/1	3	2459.38 ms	2664.83 ms	32415	12
Price First	Infron	`deepinfra`	324	98.18%	165/159	165	3793.26 ms	4203.11 ms	5776852	1296
Price First	Infron	`parasail`	6	1.82%	0/6	6	8324.74 ms	8598.95 ms	161438	24
Price First	OpenRouter	`Wafer`	312	94.55%	161/151	161	3009.99 ms	3197.16 ms	5734063	1248
Price First	OpenRouter	`DeepInfra`	7	2.12%	2/5	5	5368.65 ms	5925.45 ms	40764	28
Price First	OpenRouter	`SiliconFlow`	7	2.12%	2/5	6	6547.58 ms	6700.35 ms	105165	28
Price First	OpenRouter	`DekaLLM`	4	1.21%	0/4	4	4050.88 ms	4241.61 ms	57986	16
Latency First	Infron	`novita`	326	83.59%	167/159	184	4636.06 ms	4906.26 ms	6075350	1304
Latency First	Infron	`nextbit`	64	16.41%	28/36	53	7998.84 ms	8232.57 ms	1112281	256
Latency First	OpenRouter	`Wafer`	331	84.87%	168/163	169	2942.47 ms	3118.98 ms	6337857	1324

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Latency First	OpenRouter	`Google`	59	15.13%	27/32	33	2719.10 ms	2904.55 ms	846843	236
TTFT First	Infron	`nextbit`	267	66.75%	132/135	143	5852.33 ms	6183.27 ms	5027624	1068
TTFT First	Infron	`parasail`	78	19.50%	35/43	43	4328.70 ms	4607.51 ms	1485600	312
TTFT First	Infron	`deepinfra`	53	13.25%	32/21	32	4578.76 ms	4893.44 ms	840415	212
TTFT First	Infron	`novita`	2	0.50%	1/1	2	6630.99 ms	6687.21 ms	53834	8
TTFT First	OpenRouter	`Wafer`	327	81.75%	165/162	165	2526.51 ms	2685.14 ms	6151069	1308
TTFT First	OpenRouter	`Google`	73	18.25%	35/38	38	3186.82 ms	3464.13 ms	1253045	292

7.1 Cache-Rate and Cost Divergence Drill-Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing-mode differences.

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	-99.36 pp	1.31x	`deepinfra` 100.00%	**`Wafer` 99.19%**	+0	OpenRouter has higher cache and lower cost; provider/cache-domain mix is the main signal
Price First	-91.64 pp	1.37x	`deepinfra` 98.18%	**`Wafer` 94.55%**	+0	OpenRouter has higher cache and lower cost; provider/cache-domain mix is the main signal
Latency First	-34.75 pp	1.99x	`novita` 83.59%	**`Wafer` 84.87%**	+0	OpenRouter has higher cache and lower cost; provider/cache-domain mix is the main signal
TTFT First	-3.35 pp	1.25x	`nextbit` 66.75%	**`Wafer` 81.75%**	+0	OpenRouter has higher cache and lower cost; provider/cache-domain mix is the main signal

8. Stratified Results: Prompt-Length Cache Performance

This section aggregates second-request cache-read tokens, token-level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt-length tier. Bold cells mark the advantaged side within each tier.

Prompt-Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
`short`	1500	Infron	**250**	**522652**	184416	35.28%	$0.09358700	3671.92 ms	3335.98 ms
`short`	1500	OpenRouter	**250**	521463	**448719**	**86.05%**	**$0.06677004**	**2042.67 ms**	**1852.01 ms**
`medium`	8000	Infron	**251**	**2713228**	939376	34.62%	$0.48702400	4735.61 ms	4421.16 ms
`medium`	8000	OpenRouter	**251**	2712013	**2457340**	**90.61%**	**$0.33247589**	**3038.25 ms**	**2848.47 ms**
`long`	32000	Infron	**245**	**10537582**	3783408	35.90%	$1.86894700	5955.96 ms	5634.06 ms
`long`	32000	OpenRouter	**245**	10536427	**9576674**	**90.89%**	**$1.24177990**	**4129.40 ms**	**3963.38 ms**

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
`short`	Throughput First	0.00%	**96.58%**	**OpenRouter**
`short`	Price First	0.00%	**85.91%**	**OpenRouter**
`short`	Latency First	57.10%	**80.39%**	**OpenRouter**
`short`	TTFT First	75.20%	**82.16%**	**OpenRouter**
`medium`	Throughput First	0.00%	**97.91%**	**OpenRouter**
`medium`	Price First	1.74%	**92.53%**	**OpenRouter**
`medium`	Latency First	53.78%	**86.55%**	**OpenRouter**
`medium`	TTFT First	76.01%	**86.17%**	**OpenRouter**
`long`	Throughput First	0.00%	**99.86%**	**OpenRouter**
`long`	Price First	1.92%	**94.10%**	**OpenRouter**
`long`	Latency First	49.98%	**85.82%**	**OpenRouter**
`long`	TTFT First	83.29%	**84.73%**	**OpenRouter**

9. Stratified Results: Group-Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	48	48	0.00%	$0.12534400	16802.11 ms	16542.71 ms
Infron	2	47	47	0.00%	$0.12505000	14416.81 ms	13478.74 ms
Infron	3	47	47	0.00%	$0.12955800	10571.42 ms	10237.77 ms
Infron	4	44	44	0.00%	$0.11148600	8992.76 ms	8572.44 ms
OpenRouter	1	48	48	**99.73%**	**$0.10265040**	**7391.17 ms**	**6595.85 ms**
OpenRouter	2	47	47	**99.73%**	**$0.09459054**	**5114.44 ms**	**5055.18 ms**
OpenRouter	3	47	47	**98.58%**	**$0.09405886**	**5945.15 ms**	**5642.38 ms**
OpenRouter	4	44	44	**99.46%**	**$0.08300238**	**5575.85 ms**	**5552.48 ms**

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	40	40	0.00%	$0.10072000	9530.75 ms	8840.43 ms
Infron	2	38	38	0.00%	$0.10605600	**7790.75 ms**	**7419.75 ms**
Infron	3	42	42	0.00%	$0.10845400	14506.44 ms	13923.42 ms
Infron	4	45	45	7.41%	$0.10632500	13316.27 ms	12789.29 ms
OpenRouter	1	40	40	**98.21%**	**$0.07195559**	**6190.24 ms**	**6127.13 ms**
OpenRouter	2	38	38	**80.76%**	**$0.08333400**	8902.37 ms	8853.82 ms
OpenRouter	3	42	42	**97.28%**	**$0.07875651**	**8700.04 ms**	**8666.29 ms**
OpenRouter	4	45	45	**97.93%**	**$0.07426325**	**6595.52 ms**	**5836.52 ms**

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	47	47	63.33%	$0.22159300	13148.44 ms	12680.91 ms
Infron	2	48	48	28.09%	$0.23387100	9313.01 ms	8892.90 ms
Infron	3	50	50	49.14%	$0.23907800	9401.61 ms	9256.44 ms
Infron	4	50	50	63.93%	$0.22105000	15370.09 ms	15123.91 ms
OpenRouter	1	47	47	**72.52%**	**$0.12828628**	**5909.73 ms**	**5674.95 ms**
OpenRouter	2	48	48	**79.55%**	**$0.12797190**	**5378.67 ms**	**5352.42 ms**
OpenRouter	3	50	50	**96.36%**	**$0.10237124**	**3832.10 ms**	**3198.89 ms**

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	4	50	50	**93.30%**	**$0.10138260**	**4245.44 ms**	**4239.06 ms**

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	40.05%	$0.16318900	16516.93 ms	16009.39 ms
Infron	2	50	50	**99.94%**	$0.16964500	9934.38 ms	9793.22 ms
Infron	3	50	50	**95.14%**	$0.16183400	10216.03 ms	10076.18 ms
Infron	4	50	50	89.80%	$0.12630500	8807.96 ms	8473.19 ms
OpenRouter	1	50	50	**78.94%**	**$0.13872192**	**5094.13 ms**	**5047.26 ms**
OpenRouter	2	50	50	72.13%	**$0.15076992**	**6653.27 ms**	**6300.65 ms**
OpenRouter	3	50	50	90.42%	**$0.11096412**	**5044.54 ms**	**5024.72 ms**
OpenRouter	4	50	50	**98.55%**	**$0.09794632**	**4181.31 ms**	**3830.28 ms**

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (5/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	OpenRouter leads overall (5/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	First-token and E2E response are faster; check cache and cost
Latency First	Minimize full-response waiting time	OpenRouter leads overall (5/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	First-token and E2E response are faster; check cache and cost
TTFT First	Minimize streaming first-token time	OpenRouter leads overall (5/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	First-token and E2E response are faster; check cache and cost

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)
Latency First	Minimize full-response waiting time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)
TTFT First	Minimize streaming first-token time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (5/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	`export/gemma_4_26b_a4b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json`
Paired dataset	`export/gemma_4_26b_a4b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv`

Artifact	Path
Request-level dataset	`export/gemma_4_26b_a4b_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `benchmark_requests.jsonl`
Filtered structured records	`export/gemma_4_26b_a4b_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `records.json`
Excluded-record audit	`export/gemma_4_26b_a4b_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `records_excluded.json`
Test source	`tests/test_rerun_routing_sort_cache_cost_ab.py`
Benchmark runner source	`scripts/rerun_routing_sort_cache_cost_ab.py`
HTML report renderer source	`scripts/render_glm52_deepseek_style_report.py`
Dataset reference	`business_representative` built-in representative business templates; request-level export is `benchmark_requests`